

1. State which of the following are not the probability distributions of a random variable. Give reasons for your answer.

(i)

X	0	1	2
P(X)	0.4	0.4	0.2

(ii)

X	0	1	2	3	4
P(X)	0.1	0.5	0.2	-0.1	0.3

(iii)

Y	-1	0	1
P(Y)	0.6	0.1	0.2

(iv)

Z	3	2	1	0	-1
P(Z)	0.3	0.2	0.4	0.1	0.05

2. Find the probability distribution of

- (i) number of heads in two tosses of a coin
(ii) number of tails in the simultaneous tosses of three coins
(iii) number of heads in four tosses of a coin

3. A coin is biased so that the head is 3 times as likely to occur as tail. If the coin is tossed twice, find the probability distribution of number of tails.

4. A random variable X has the following probability distribution.

X	0	1	2	3	4	5	6	7
P(X)	0	k	2k	2k	3k	k ²	2k ²	7k ² + k

Determine i) k ii) $P(X < 3)$ iii) $P(X > 6)$ iv) $P(0 < x < 3)$

5. The random variable X has probability distribution P(X) of the following form, where k is some number:

$$P(X) = \begin{cases} k, & \text{if } x = 0 \\ 2k, & \text{if } x = 1 \\ 3k, & \text{if } x = 2 \\ 0, & \text{otherwise} \end{cases}$$

(a) Determine the value of k.

(b) Find $P(X < 2)$, $P(X \geq 2)$, $P(X \geq 2)$.

6. Let X be a discrete random variable with a probability function

Values of X : x	1	4	9
f(x)	0.1	0.4	0.5

Find the mean of X, X^2 , $\sqrt{X}$ and $2\sqrt{X}$.